Andrea Amaduzzi

PhD in Computer Vision & Multimodal AI · 3D-Language Understanding · Large-Scale LLM Training

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Research Experience

PhD – Multimodal AI for 3D-Language Understanding — Nov 2021 – March 2026
Computer Vision Laboratory, University of Bologna, Italy
Advisor: Prof. Luigi Di Stefano

- Published 3 first-author papers at top-tier ML/CV conferences (NeurIPS, ICCV); 1 first-author journal submission at TPAMI.
- Designed and fine-tuned multimodal LLMs for 3D scene understanding, integrating NeRFs, point clouds, and language.
- Led large-scale distributed training of 10B+ parameter models on LEONARDO, a top-10 global supercomputer, using DeepSpeed and FSDP.
- Built custom datasets and benchmarks for evaluating 3D-language alignment, grounding, and generation quality.

Niantic Spatial — London, UK
Research Intern — Sept 2025 – March 2026

- Designed and fine-tuned multimodal LLMs (10B+ parameters) for outdoor metric spatial reasoning.
- Created the first data annotation pipeline for outdoor spatial reasoning, using images, camera poses and depth maps as input modalities.
- Built a purpose-designed MLLM that achieves state-of-the-art on the proposed benchmark.
- Advisors: Tommaso Cavallari and Victor Adrian Prisacariu.

KUKA – Corporate Research — Augsburg, Germany
Research Intern — July 2019 – Jan 2020

- Built deep learning pipelines for semantic segmentation and action recognition using multimodal sensor data.
- Mentored 30+ students on deep learning workflows and 3D sensor projects.

Industry Experience

Datalogic — Bologna, Italy
R&D Vision Software Engineer — June 2020 – Nov 2021

- Developed and maintained production-grade Python and C++ ML libraries, deployed on millions of edge devices.
- Optimized ML pipelines under real-world constraints, reducing production error rates by $>$50%.

Publications

Andrea Amaduzzi, Pierluigi Zama Ramirez, Giuseppe Lisanti, Samuele Salti, Luigi Di Stefano
"Spatially-aware Weights Tokenization for NeRF-Language Models"
NeurIPS 2025 (top-tier ML conference)

Andrea Amaduzzi, Pierluigi Zama Ramirez, Giuseppe Lisanti, Samuele Salti, Luigi Di Stefano
"Scaling LLaNA: Advancing NeRF-Language Understanding Through Large-Scale Training"
under review at IEEE TPAMI (top-tier CV journal)

Andrea Amaduzzi, Pierluigi Zama Ramirez, Giuseppe Lisanti, Samuele Salti, Luigi Di Stefano
"LLaNA: Large Language and NeRF Assistant"
NeurIPS 2024 (top-tier ML conference)

Andrea Amaduzzi, Giuseppe Lisanti, Samuele Salti, Luigi Di Stefano
"Looking at words and points with attention: a benchmark for text-to-shape coherence"
ICCV 2023 Workshop (top-tier CV conference workshop)

Presentations

Poster: "Spatially-aware Weights Tokenization for NeRF-Language Models" — NeurIPS 2025, Vancouver

Invited Talk: "LLaNA: Large Language and NeRF Assistant" — CINECA AI User Forum, Bologna
HPC User Day – Focus on Leonardo supercomputer — March 2025

Poster: "LLaNA: Large Language and NeRF Assistant" — NeurIPS 2024, Vancouver

Poster: "Looking at words and points with attention" — ICCV Workshop 2023, Paris

Education

M.S. in Automation Engineering — Sept 2017 – March 2020
University of Bologna, Italy

- Visiting researcher at the University of Technology of Sydney (GPA: 3.88/4.0, Top 1%).
- Thesis on deep learning for human action recognition, at KUKA Corporate Research, Augsburg, Germany.
- Graduation: 110/110 cum laude. GPA: 29.23/30.00 (Top 3%).

B.S. in Automation Engineering — Sept 2014 – July 2017
University of Bologna, Italy

- Graduation: 110/110 cum laude. GPA: 28.71/30.00 (Top 1%).

Teaching

Teaching Assistant at University of Bologna for graduate and undergraduate courses:
Computer Vision and Image Processing M (Fall 2024), Logic Design T (Spring 2022, 2023, 2024).

Technical Skills

Research Areas: Multimodal LLMs, Vision-Language Models, 3D Scene Understanding, Neural Radiance Fields, Spatial Reasoning, Generative Diffusion Models.

Frameworks & Infrastructure: PyTorch, DeepSpeed, FSDP, Hugging Face Transformers, distributed multi-node GPU training, SLURM.

Languages & Tools: Python, C++, C, Git, Linux, Bash, LaTeX, JavaScript.

Spoken Languages: Italian (native), English (IELTS 8.0), German (A2), Spanish (A2).